JAVA

Java is a high-level, object-oriented, platform-independent programming language developed by **Sun Microsystems** (now owned by **Oracle**).

James Gosling invented the java programming language.

It is widely used for developing web applications, mobile apps, desktop software, enterprise systems, and games.

Features of Java

- **Platform Independent:** Java programs run on any operating system using the Java Virtual Machine (JVM). This is known as **"Write Once, Run Anywhere (WORA)."**
- **Object-Oriented:** Java is based on objects and classes, making code reusable and easy to maintain.
- **Simple:** Java has a clear and easy-to-learn syntax.
- **Secure:** It provides built-in security features such as bytecode verification and exception handling.
- **Robust:** Strong memory management and error handling make Java reliable.
- **Multithreaded:** Java allows multiple tasks to run simultaneously.
- **Portable:** Java programs can be transferred and executed on different platforms without modification.

Java Architecture

Java Source Code (.java)

↓

Java Compiler (javac)

↓

Bytecode (.class)

↓

Java Virtual Machine (JVM)

↓

Program Output

Basic Java Program

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

Explanation

- ✓ public class HelloWorld – Defines a class named HelloWorld.
- ✓ public static void main(String[] args) – The main method where program execution starts.
- ✓ System.out.println() – Prints text to the console.

Applications of Java

- Web applications
- Android applications
- Desktop applications
- Enterprise software
- Banking systems
- Cloud-based applications
- Scientific applications

JIT (Just-In-Time) Compiler

Definition

The **JIT (Just-In-Time) Compiler** is a component of the **Java Virtual Machine (JVM)** that improves the execution speed of Java programs by converting frequently used **bytecode** into **native machine code** at runtime.

Step-by-Step Process

1. Java source code (.java) is compiled into **bytecode (.class)**.
2. The JVM loads the bytecode.
3. Initially, the **interpreter** executes the bytecode line by line.
4. The JVM monitors which methods are executed frequently (called **hot spots**).
5. The **JIT compiler** compiles these frequently used methods into native machine code.
6. The compiled machine code is stored in memory.
7. Subsequent calls execute the native code directly, improving performance.

Example

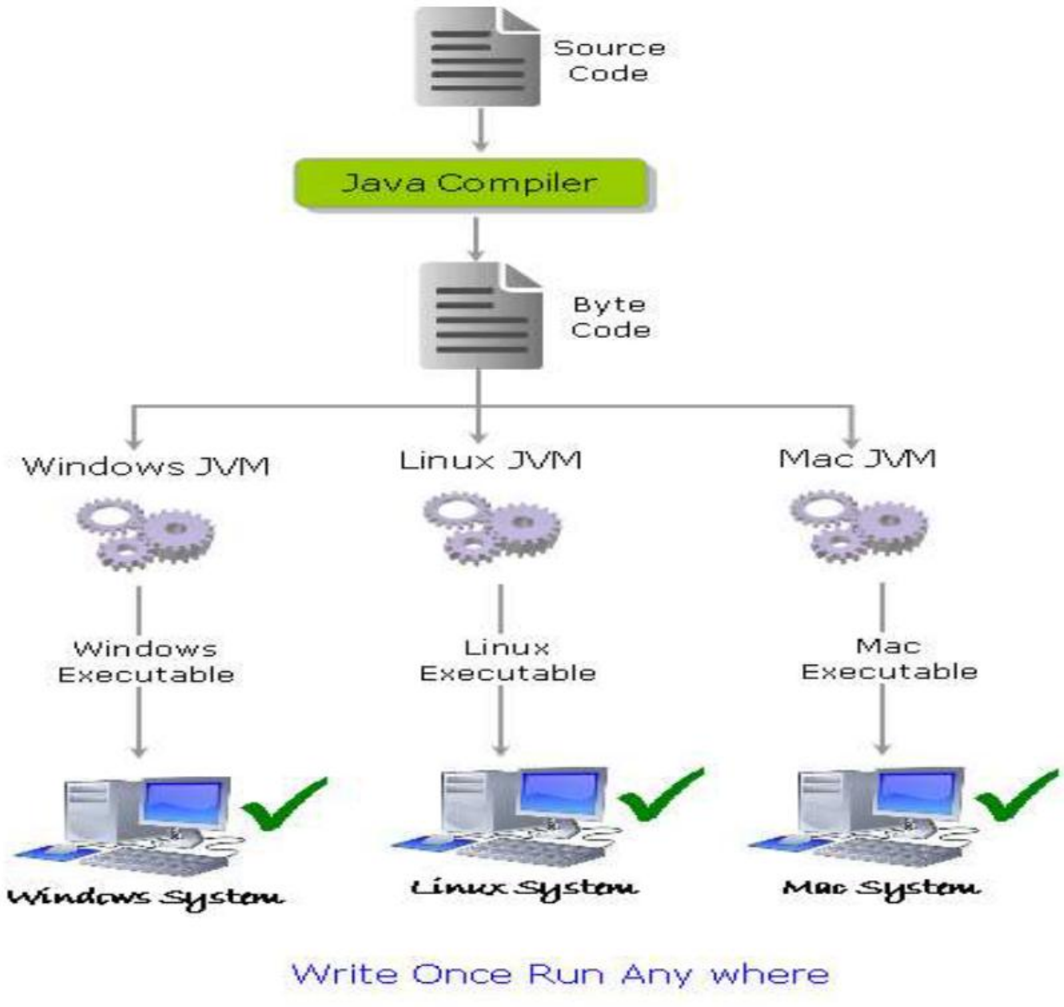
```
public class JITExample {
    public static void main(String[] args) {
        long sum = 0;
        for (int i = 0; i < 100000000; i++) {
            sum += i;
        }
        System.out.println(sum);
    }
}
```

Explanation

- The for loop executes millions of times.
- Initially, the JVM interprets the bytecode.
- Since the loop becomes a **hot spot**, the JIT compiler converts it into native machine code.
- The remaining iterations run much faster.

Java Virtual Machine (JVM)

- A Virtual Machine is a software implementation of a physical machine. Java was developed with the concept of WORA (Write Once Run Anywhere), which runs on a VM.
- The compiler compiles the Java file into a Java .class file, then that .class file is input into the JVM, which loads and executes the class file.



Types of Java Programs

Stand-alone applications (J2SE)

AWT (Abstract Window Toolkit) and swing are used to create stand-alone application.

Ex: Student Management System, Library Management System

Web applications (JSP)

Applet, Servlet, Java Server Pages (JSP), Java Server Faces (JSF) which are used to create web applications

Ex: Online Shopping Website, Student login, View attendance, Submit assignments

Enterprise applications (J2EE)

Designed for corporate sides such as Banking business systems, Hospital Management System, Airline Reservation System

Mobile applications (J2ME)

Mobile application is developed for run the mobile

Java Applet

A Java Applet is a small Java program that runs inside a web browser or Applet Viewer to provide interactive features such as graphics, animations, and user interaction.

Applet Viewer is a JDK tool used to execute and test Java applets without using a web browser.

Java Command Line Arguments

String[] → An array of String objects.

args → Variable name (you can use any valid name, not necessarily args)

Print all command-line arguments

```
class Test {
    public static void main(String[] args) {
        System.out.println("Number of arguments: " + args.length);
        for (int i = 0; i < args.length; i++) {
            System.out.println("Argument " + i + " = " + args[i]);
        }
    }
}
```

javac Test.java

java Test Apple Banana Orange

Add two numbers using command-line arguments

```
class AddNumbers {
    public static void main(String[] args) {
```

```
        int a = Integer.parseInt(args[0]);
        int b = Integer.parseInt(args[1]);
        System.out.println("Sum = " + (a + b));
    }
}
```

```
javac AddNumbers.java
java AddNumbers 25 35
```

Reading Input from the user

```
import java.lang.*;
import java.util.Scanner;
class Hai
{
public static void main(String[] args)
{
Scanner s=new Scanner(System.in);
int a=s.nextInt();
int b=s.nextInt();
System.out.print("sum="+(a+b));}
}
```